08. Coordinates & Transformations

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Reading coordinates:

A point A is at (5, 2) on a coordinate grid. What is its x-coordinate and what is its y-coordinate?

Answer: The x-coordinate is given first: **5**. The y-coordinate is given second: **2**.

Example 2 — Translating a point:

A point B is at (3, 6). It is translated 4 right and 2 down. What are its new coordinates?

Answer: 4 right: 3 + 4 = 7. 2 down: 6 − 2 = 4. New point: **(7, 4)**.

Example 3 — Reflecting in the x-axis:

A point C is at (6, 5). It is reflected in the x-axis. What are its new coordinates?

Answer: Reflecting in the x-axis keeps x the same and reverses the sign of y: **(6, -5)**.

Example 4 — Reflecting in the y-axis:

A point D is at (-3, 4). It is reflected in the y-axis. What are its new coordinates?

Answer: Reflecting in the y-axis keeps y the same and reverses the sign of x: **(3, 4)**.

Example 5 — Multi-step: translate, then reflect:

A point E is at (2, 3). It is translated 5 right and 1 up to give point F. Point F is then reflected in the x-axis. What are the final coordinates?

Answer: First translate: (2 + 5, 3 + 1) = F(7, 4). Then reflect F in the x-axis (keep x, reverse sign of y): **(7, -4)**.

Foundation

15 questions

1. A point is at (4, 7). What is its x-coordinate?

2. A point is at (4, 7). What is its y-coordinate?

3. A point starts at (2, 3) and is translated 4 right. What are its new coordinates?

4. A point starts at (5, 2) and is translated 3 up. What are its new coordinates?

5. A point is at (1, 4). It is translated 3 right. What are its new coordinates? A) (4, 4) B) (1, 7) C) (4, 1) D) (1, 1)

6. Point B is at (0, 5). Which axis does it lie on?

7. Point C is at (7, 0). Which axis does it lie on?

8. A point starts at (3, 5) and is translated 2 right and 3 up. What are its new coordinates?

9. A point starts at (6, 6) and is translated 5 left. What are its new coordinates?

10. Select ALL that apply. Look at the points P(4, 4), Q(4, 8), R(9, 4) and S(9, 8). Which have an x-coordinate of 4? A) P(4, 4) B) Q(4, 8) C) R(9, 4) D) S(9, 8)

11. A point starts at (0, 0) and is translated 7 right and 3 up. What are its new coordinates?

12. A point starts at (5, 5) and is translated 4 down. What are its new coordinates? A) (5, 1) B) (1, 5) C) (9, 5) D) (5, 9)

13. A point D is at (3, 3) and a point E is at (3, 9). Are D and E on the same vertical line or the same horizontal line?

14. A point starts at (6, 2) and is translated 3 right and 4 up. What are its new coordinates?

15. Which of these points lies exactly on the x-axis? A) (5, 0) B) (0, 7) C) (3, 3) D) (6, 6)

Standard

20 questions

1. A point starts at (4, 5) and is translated 6 right and 3 up. What are its new coordinates?

2. A point is at (5, 4). It is reflected in the x-axis. What are its new coordinates?

3. A point starts at (6, 7) and is translated 5 down. What are its new coordinates?

4. A point at (3, 7) is reflected in the x-axis. What are its new coordinates? A) (3, -7) B) (-3, 7) C) (7, 3) D) (-3, -7)

5. A point is at (4, 6). It is reflected in the y-axis. What are its new coordinates?

6. A rectangle has vertices at (2, 2), (7, 2), (7, 5) and (2, 5). What is the length of its longer side?

7. A point starts at (7, 3) and is translated 2 right and 6 up. What are its new coordinates?

8. A point starts at (5, 9) and is translated 7 left. What are its new coordinates?

9. A point at (6, 3) is reflected in the y-axis. What are its new coordinates? A) (-6, 3) B) (6, -3) C) (3, 6) D) (-6, -3)

10. A point starts at (0, 8) and is translated 4 right and 3 down. What are its new coordinates?

11. Select ALL that apply. A square has vertices at (3, 3), (3, 7), (7, 7) and (7, 3). Which statements are true? A) The square has a side length of 4 units B) The square has a side length of 7 units C) The point (7, 7) is a vertex of the square D) The point (3, 7) is a vertex of the square

12. A point starts at (3, 4) and is translated 5 right and 5 up. What are its new coordinates?

13. A point is at (8, 2). It is reflected in the x-axis. What are its new coordinates?

14. A point starts at (4, 4) and is translated 3 left and 6 up. What are its new coordinates? A) (1, 10) B) (7, 10) C) (1, -2) D) (7, -2)

15. A point starts at (9, 0) and is translated 3 up. What are its new coordinates?

16. A point starts at (5, 5) and is translated 2 right and 2 down. What are its new coordinates?

17. A rectangle has vertices at (1, 1), (1, 6), (8, 6) and (8, 1). What is the length of its shorter side?

18. A point is at (9, 4). It is reflected in the y-axis. What are its new coordinates?

19. A point starts at (2, 2) and is translated 6 right and 6 up. What are its new coordinates?

20. Select ALL of the following statements that are TRUE: A) Reflecting a point in the x-axis keeps the x-coordinate the same B) Reflecting a point in the y-axis keeps the y-coordinate the same C) Translating a point right increases its x-coordinate D) Translating a point down increases its y-coordinate

Hard

5 questions

1. A point starts at (5, 3). It is translated 4 right and 2 up to give point P. Point P is then reflected in the x-axis. What are the final coordinates?

2. A rectangle has vertices at (0, 0), (9, 0), (9, 5) and (0, 5). What are the coordinates of the vertex diagonally opposite (0, 0)?

3. A point Q is at (6, -2). It is reflected in the y-axis to give point R. Point R is then translated 3 left and 4 up. What are the final coordinates?

4. A square has one vertex at (2, 5) and sides parallel to the axes. Its vertex diagonally opposite is at (8, 11). Select ALL the statements below that are TRUE: A) The square's other two vertices are (2, 11) and (8, 5) B) The side length is 6 units C) The perimeter of the square is 24 units D) The area of the square is 42 square units

5. A point starts at (-4, 6). It is translated 7 right and 5 down to give point M. Point M is then reflected in the x-axis. What are the final coordinates?

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